# INFORMATION PRACTICES (IP)

Maximum Marks: 100

Theory: Marks 70.

Practicals: Marks 30. External: 20 marks, Internal: 10 marks

| TOPIC | Marks | Theory Lectures | Practical |
|---|---|---|---|
| Basics of Information Technology | 20 | 40 | |
| Operating System Concepts | 10 | 30 | - |
| Programming Concepts | 10 | 20 | |
| Python Programming | 20 | 35 | 30 |
| Introduction to Emerging Technologies | 10 | 25 | |

**UNIT –I Basics of Information Technology**

Introduction to Information Technology, Data, Information, Importance of Information Technology, Components of Information Technology.

Introduction to Computer: Basic components of a Computer System (CPU, ALU, CU, Memory) and their interconnection, Illustration with Block Diagram, Basic Input and Output devices, Computer Memory and Types of Memory, Computer Storage and Types of Computer Storage, Units of Memory.

Concept of Software, Categories of Software (Open and Closed Source Software), Types of Software (Application and System Software)

Number Systems and Logical Gates: Binary, Octal, Decimal, Hexadecimal and conversions, Basic Logical Gates (AND, OR, NOT) with Truth Tables.

**UNIT –II Operating System Concepts**

Operating System, Types of Operating System: Batch Operating System Multitasking/Time Sharing Operating System, Multiprocessing Operating System, Real Time Operating System, Distributed Operating System, Network Operating System, Mobile Operating System.

Functions of Operating System: Processor Management, Memory Management, File Management, Device Management. 32-Bit and 64-Bit Operating System. Introduction to Firmware.

Introduction to Microsoft Windows Operating System, Features of Windows Operating System, Versions of Windows Operating System, File structure of Windows Operating System.

**UNIT –III Programming Concepts**

Introduction to Programming Languages, Categories of Computer Languages: Low Level Language, High Level Languages, Assembly Language, Language Translators: Compiler, Interpreter and Assembler, History of Programming Languages.

Basic Element of Programming Language, Syntax and Semantics of a Programming Language, Introduction to Algorithms, Pseudo code and Flowcharts.

## UNIT –IV Python Programming

Basics of Python Programming: Installation and Setup of Python and IDE, Identifiers, Keywords, Constants, Variables, Naming rules of variables, Operators (Arithmetic, Assignment, Comparison, Logical, Identity, Membership and Bitwise), Data Types, Indentation, Statements, Expressions, Input and Output Statements, Control Statement (conditional/selection statements) IF, IF-ELSE, IF-ELIF-ELSE, Looping Structure while Loop and For Loop

## UNIT –V Introduction to Emerging Technologies

Big Data, Artificial Intelligence, Machine Learning, Natural Language Processing, Immersive Technologies (Augmented Reality, Virtual Reality), Robotics, Internet of Things (IoT), Cloud Computing and Types of Cloud Services.

### Scheme of Assessment (Theory)

The Question paper shall contain - Four sections A, B, C and D. Each section is compulsory.

1. Section A-Question 1 to 10 comprises of 10 questions of 1 mark each (MCQ's, Fill in the blanks, True/False, Assertion-Reason etc.)
2. Section B-Question 11 to 19 comprises of 9 Very Short Answer (VSA)-type questions of 2 marks each.
3. Section C-Question 20 to 28 comprises of 9 Short Answer (SA)-type questions of 3 marks each.
4. Section D-Question 29 to 31 comprises of 3 Long Answer (LSA)-type questions of 5 marks each with internal choice

| PATTERN | MARKS | QUESTIONS | TOTAL |
|---|---|---|---|
| Very Short Questions | 01 | 10 | 10 Marks |
| Short Type-I Questions | 02 | 9 | 18 Marks |
| Short Type-II Questions | 03 | 9 | 27 Marks |
| Long Type Questions | 05 | 3 | 15 Marks |
| TOTAL | | 31 | 70 Marks |

## PRACTICALS

List of Suggested Practical (Programming in Python)

1. To do basic arithmetic operations.
2. To calculate Simple and Compound Interest.
3. To calculate perimeter/circumference and area of shapes such as triangle, rectangle, square and circle.
4. To find if a number is Even or Odd
5. To find if a number is Prime or Not.
6. To find Total Marks, Percentage and Grade for a given student.
7. To Swap the Two Numbers.
8. To print the multiplication table of a given number.
9. To find the sum of 'n' natural numbers.
10. To find the factorial of a natural number.